

Rutherford's Atomic Model

1. When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze' its distance of closest approach from the nucleus depends on m as:

(a) $1/m$ (b) $1/\sqrt{m}$ (c) $1/m^2$ (d) m

2. An alpha nucleus of energy $\left(\frac{1}{2}mv^2\right)$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to

(a) $1/Ze$ (b) v^2 (c) $1/m$ (d) $1/v^4$

3. In Rutherford scattering experiment, what will be the correct angle for α-scattering for an impact parameter, $b = 0$?

(a) 90° (b) 280° (c) 0° (d) 180°

4. In Rutherford scattering experiment when a projectile of charge Z_1 and mass M_1 approaches a target nucleus of charge Z_2 and mass M_2 the distance to closest approached is r_0. The energy of the projectile is

(a) Directly proportional to Z_1Z_2
(b) Inversely proportional to Z_1
(c) Directly proportional to mass M_1
(d) Directly proportional to $M_1 \times M_2$

5. Choose the correct option from the following options given below:

(a) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.
(b) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model
(c) A classical atom based on Rutherford's model is doomed to collapse
(d) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

6. Given below are two statements:

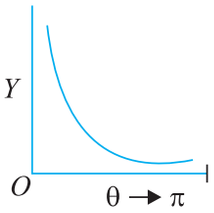
Statement I: Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

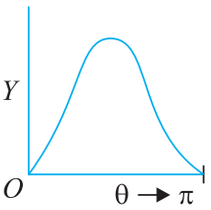
Statement II: An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model. In the light of the above statements, choose the most appropriate from the options given below.

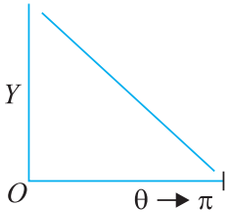
(a) Both Statement I and Statement II are false
(b) Statement I is false but Statement II is true
(c) Statement I is true but Statement II is false
(d) Both Statement I and Statement II are true

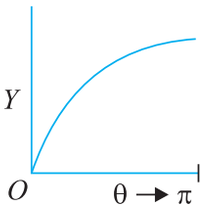
7. The graph which depicts the results of Rutherford's gold foil experiment with α-particles is

θ : Scattering angle
Y : Number of scattering a-particles detected
(Plots are schematic and not to scale)

(a)


(b)


(c)


(d)


Bohr's Atomic Model

8. The Bohr model of atoms

(a) Assumes that the angular momentum of electrons is quantized.
(b) Uses Einstein's photoelectric equation.
(c) Predicts continuous emission spectra for atoms
(d) Predicts the same emission spectra for all types of atoms.

9. In terms of Bohr radius the a_0 the second Bohr orbit of a hydrogen atom is given by

(*a*) $4a_0$ (*b*) $9a_0$ (*c*) $\sqrt{2}a_0$ (*d*) $2a_0$

10. When hydrogen is in its first excited level, its radius is..... of the Bohr radius.

(*a*) Twice (*b*) 4 times (*c*) Same (*d*) Half

11. The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} s. The frequency of revolution of the electron in its first excited state (in s^{-1}) is:

(*a*) 6.2×10^{15} (*b*) 5.6×10^{12}

(*c*) 7.8×10^{14} (*d*) 1.6×10^{14}

12. The speed of an electron in the 4th orbit of hydrogen atom is

(*a*) c (*b*) $\frac{c}{137}$ (*c*) $\frac{c}{2192}$ (*d*) $\frac{c}{548}$

13. The angular momentum of an electron in a hydrogen atom is proportional to (where r is radius of orbit).

(*a*) $\frac{1}{\sqrt{r}}$ (*b*) $\frac{1}{r}$ (*c*) $\sqrt{r}$ (*d*) r^2

14. What should be the angular momentum of an electron in Bohr's hydrogen atom whose energy is –0.544 eV?

(*a*) $\frac{h}{\pi}$ (*b*) $\frac{3h}{2\pi}$ (*c*) $\frac{5h}{2\pi}$ (*d*) $\frac{2h}{2\pi}$

15. According to Bohr's theory, the angular momentum of an electron revolving in 4th orbit of hydrogen atom is:

(*a*) $8\frac{h}{\pi}$ (*b*) $\frac{h}{\pi}$ (*c*) $2\frac{h}{\pi}$ (*d*) $\frac{h}{2\pi}$

16. In Bohr's model of the hydrogen atom, the ratio between the period of revolution of an electron in the orbit of $n = 1$ to the period of revolution of the electron in the orbit $n = 2$ is

(*a*) 2 : 1 (*b*) 1 : 2 (*c*) 1 : 4 (*d*) 1 : 8

17. The magnetic field of induction produced at the centre of orbit due to an electron revolving in n^{th} orbit of hydrogen atom is proportional to

(*a*) n^{-3} (*b*) n^{-5} (*c*) n^5 (*d*) n^3

18. In which of the following systems will be radius of the first orbit ($n = 1$) be minimum?

(*a*) Doubly ionized lithium (*b*) Singly ionized helium

(*c*) Deuterium atom (*d*) Hydrogen atom

19. The energy of hydrogen atom in n^{th} orbit is E_n then the energy in n^{th} orbit of singly ionized helium atom will be

(*a*) $4E_n$ (*b*) $E_n/4$ (*c*) $2E_n$ (*d*) $E_n/2$

20. Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

(*a*) 6 (*b*) 8 (*c*) 4 (*d*) 1

21. In the Bohr model of a hydrogen atom, the centripetal force is furnished by the coulomb attraction between the proton and the electron. If a_0 is the radius of the ground state orbit, m is the mass and e is the charge on the electron and ε_0 is the vacuum permittivity, the speed of the electron is

(*a*) $\frac{e}{\sqrt{4\pi\varepsilon_0 a_0 m}}$ (*b*) $\frac{e}{\sqrt{\varepsilon_0 a_0 m}}$

(*c*) 0 (*d*) $\frac{\sqrt{4\pi\varepsilon_0 a_0 m}}{e}$

22. An electron of a stationary hydrogen atom passes from the fifth energy level to the ground level. The velocity that the atom acquired as a result of photon emission will be

(*a*) $\frac{24hR}{25m}$ (*b*) $\frac{25hR}{24m}$

(*c*) $\frac{25m}{24hR}$ (*d*) $\frac{24m}{25hR}$

(m is the mass of the electron, R is Rydberg constant and h is Planck's constant)

23. A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2} m\omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of angular momentum and circular orbit, the radius of nth orbit will be proportional to:

(*a*) $\sqrt{n}$ (*b*) n (*c*) n^2 (*d*) $1/n$

24. In Bohr model of hydrogen atom, linear momentum of the electron depends on the principal quantum number n as

(*a*) $p_n \propto \frac{1}{n}$ (*b*) $p_n \propto n$

(*c*) $p_n \propto n^2$ (*d*) $p_n \propto 1/n^2$

25. In Bohr model of hydrogen atom, the angular velocity of the electron depends on the principal quantum number as

(*a*) $\omega_n \propto n^2$ (*b*) $\omega_n \propto n^3$

(*c*) $\omega_n \propto 1/n^3$ (*d*) $\omega_n \propto 1/n^2$

26. The radius of 2nd orbit of He^+ of Bohr's model is r_1 and that of fourth orbit of Be^{3+} is represented as r_2. Now the ratio r_2/r_1 is $x : 1$. The value of x is _____.

27. The radius of fifth orbit of the Li^{++} is _____ $\times 10^{-12}$ m.
Take : radius of hydrogen atom = 0.51 Å.

28. The momentum of an electron revolving in n^{th} orbit is given by: (Symbols have their usual meanings)

(*a*) $\frac{nh}{2\pi r}$ (*b*) $\frac{nh}{2r}$ (*c*) $\frac{nh}{2\pi}$ (*d*) $\frac{2\pi r}{nh}$

29. In Bohr model of hydrogen atom, the force on the electron depends on the principal quantum number n as

(*a*) $F \propto 1/n^3$ (*b*) $F \propto 1/n^4$

(*c*) $F \propto 1/n^5$ (*d*) does not depend on n

30. The magnetic moment (M) of a revolving electron around the nucleus varies with principal quantum number (n) as

(*a*) $M \propto n$ (*b*) $M \propto \frac{1}{n}$

(*c*) $M \propto n^2$ (*d*) $M \propto \frac{1}{n^2}$

31. The radius of the smallest electron orbit in hydrogen-like ion is $\frac{0.51}{4}\times10^{-10}$ m, then it is

(*a*) hydrogen atom (*b*) H^+
(*c*) Li^{2+} (*d*) Be^{3+}

Kinetic Energy and Potential Energy

32. In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level.

(*a*) All K, P and E increase
(*b*) K decreases, P and E increase
(*c*) P decreases, K and E increase
(*d*) K increases, P and E decrease

33. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is

(*a*) 1 : 1 (*b*) 1 : –1 (*c*) 2 : –1 (*d*) 1 : –2

34. The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is:

(*a*) 4 (*b*) $\frac{1}{4}$ (*c*) $\frac{1}{2}$ (*d*) 1

35. An electron rotates in a circle around a nucleus having positive charge *Ze*. Correct relation between total energy (*E*) of electron to its potential energy (*U*) is:

(*a*) $E = 2U$ (*b*) $2E = 3U$ (*c*) $E = U$ (*d*) $2E = U$

Hydrogen Spectrum

36. Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is

(*a*) $\frac{7}{29}$ (*b*) $\frac{9}{31}$ (*c*) $\frac{5}{27}$ (*d*) $\frac{5}{23}$

37. Electron in hydrogen atom first jumps from third excited state to second excited state and then from second excited to the first excited state. The ratio of the wavelengths λ_1 : λ_2 emitted in the two cases is

(*a*) $\frac{7}{5}$ (*b*) $\frac{27}{20}$ (*c*) $\frac{27}{5}$ (*d*) $\frac{20}{7}$

38. The ratio of wavelengths of the last line of balmer series and the last line of Lyman series is

(*a*) 1 (*b*) 4 (*c*) 0.5 (*d*) 2

39. The energies of three conservative energy levels L_3, L_2 and L_1 of hydrogen atom are E_0, $\frac{4E_0}{9}$ and $\frac{E_0}{4}$ respectively. A photon of wavelength λ is emitted for a transition L_3 to L_1. What will be the wavelength of emission for transition L_2 to L_1?

(*a*) $\frac{16\lambda}{31}$ (*b*) $\frac{27\lambda}{7}$ (*c*) $\frac{19}{20}\lambda$ (*d*) λ

40. When an electron is excited to n^{th} energy state in hydrogen, the possible of spectral lines emitted are

(*a*) n (*b*) $2n$
(*c*) $\frac{n^2-n}{2}$ (*d*) $\frac{n^2+n}{2}$

41. In which transition of a hydrogen, atom, photons of lowest frequency are emitted?

(*a*) $n = 4$ to $n = 3$ (*b*) $n = 4$ to $n = 2$
(*c*) $n = 2$ to $n = 1$ (*d*) $n = 3$ to $n = 1$

42. Given the value of Rydberg constant is 10^7 m^{-1}, the wave number of the last line of the Balmer series in hydrogen spectrum will be

(*a*) 0.25×10^7 m^{-1} (*b*) 2.5×10^7 m^{-1}
(*c*) 0.025×10^4 m^{-1} (*d*) 0.5×10^7 m^{-1}

43. Three energy levels L_1, L_2 and L_3 of a hydrogen atom correspond to increasing values of energy *i.e.*, $E_{L1} < E_{L2} < E_{L3}$. If the wavelength corresponding to the transitions L_3 to L_2, L_2 to L_1, and L_3 to L_1, are λ_3, λ_2 and λ_1 respectively then

(*a*) $\lambda_3 = \lambda_1 + \lambda_2$ (*b*) $\lambda_1 = \frac{\lambda_2\lambda_3}{\lambda_2+\lambda_3}$
(*c*) $\lambda_1 = \lambda_2 + \lambda_3$ (*d*) $\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

44. An electron makes a transition from orbit $n = 4$ to the orbit $n = 2$ of a hydrogen atom. What is the wavelength of the emitted radiations? (R = Rydberg's constant)

(*a*) $\frac{16}{4R}$ (*b*) $\frac{16}{5R}$ (*c*) $\frac{16}{2R}$ (*d*) $\frac{16}{3R}$

45. If Ryderberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where α = _____.

46. The wavelength of the first line of Lyman series for hydrogen atom is equal to that of the second line of Balmer series for a hydrogen like ion. The atomic number Z of hydrogen like ion is

(*a*) 3 (*b*) 4 (*c*) 1 (*d*) 2

47. Consider 3rd orbit of He^+ (Helium), using non-relativistic approach, the speed of electron in this orbit will be [given K = 9×10^9 constant, Z =2 and h (Planck's constant) = 6.6×10^{-34} J s]

(*a*) 0.73×10^6 m/s (*b*) 3.0×10^8 m/s
(*c*) 292×10^6 m/s (*d*) 1.46×10^6 m/s

48. An electron in hydrogen atom makes a transition $n_1 \rightarrow n_2$ where n_1 and n_2 are principal quantum numbers of the two states. Assuming Bohr's model to be valid, the time period of the electron in the inital state is eight times that in the final state. The possible values of n_1 and n_2 are

(*a*) $n_1 = 6$ and $n_2 = 2$ (*b*) $n_1 = 8$ and $n_2 = 1$
(*c*) $n_1 = 8$ and $n_2 = 2$ (*d*) $n_1 = 4$ and $n_2 = 2$

49. The energy of a hydrogen atom in the ground state is – 13.6 eV. The energy of a He^+ ion in the first excited state will be

(*a*) – 13.6 eV (*b*) – 27.2 eV
(*c*) – 54.4 eV (*d*) – 6.8 eV

50. The total energy of electron in the ground state of hydrogen atom is – 13.6 eV. The kinetic energy of an electron in the first excited state is

(*a*) 6.8 eV (*b*) 13.6 eV (*c*) 1.7 eV (*d*) 3.4 eV

51. The transition from the state $n = 3$ to $n = 1$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from

(*a*) $2 \rightarrow 1$ (*b*) $3 \rightarrow 2$ (*c*) $4 \rightarrow 2$ (*d*) $4 \rightarrow 3$

52. Hydrogen atom in ground state is excited by monochromatic radiation of $\lambda = 975$ Å. Number of spectral lines in the resulting spectrum emitted will be

(*a*) 3 (*b*) 2 (*c*) 6 (*d*) 10

53. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be

(*a*) $\frac{16}{25}\lambda$ (*b*) $\frac{9}{16}\lambda$ (*c*) $\frac{20}{7}\lambda$ (*d*) $\frac{20}{13}\lambda$

54. A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of:
(Given, Planck's constant = 6.6×10^{-34} Js)

(*a*) 2.10×10^{-34} Js (*b*) 1.05×10^{-34} Js
(*c*) 3.15×10^{-34} Js (*d*) 4.2×10^{-34} Js

55. The recoil speed of a hydrogen atom after it emits a photon in going from $n = 5$ state to $n = 1$ state will be:

(*a*) 4.17 m/s (*b*) 2.19 m/s (*c*) 4.34 m/s (*d*) 3.25 m/s

56. In a hydrogen like atom, when an electron jumps from the M-shell to the L-shell the wavelength of the emitted radiation is λ. If an electron jumps from N-shell to the L-shell the wavelength of emitted radiation will be:

(*a*) $\frac{27}{20}\lambda$ (*b*) $\frac{16}{25}\lambda$ (*c*) $\frac{25}{16}\lambda$ (*d*) $\frac{20}{27}\lambda$

57. The minimum energy required by a hydrogen atom in ground state to emit radiation in Balmer series is nearly:

(*a*) 1.5 eV (*b*) 13.6 eV (*c*) 1.9 eV (*d*) 12.1 eV

58. In Franck-Hertz experiment, the first dip in the current-voltage graph for hydrogen is observed at 10.2 eV. The wavelength of light emitted by hydrogen atom when excited to the first excitation level is _____ nm.

(Given $hc = 1245$ eV nm, $e = 1.6 \times 10^{-19}$ C)

59. $\frac{x}{x+4}$ is the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its:

(*i*) Third permitted energy level to the second level and
(*ii*) The highest permitted energy level to the second permitted level

The value of x will be _____.

60. In hydrogen atom, electron making transition between different state.

Column I Initial and final state		Column II No. of unique wavelengths	
(*a*)	$n_2 = 4$ to $n_1 = 1$	(*i*)	10
(*b*)	$n_1 = 5$ to $n_1 = 2$	(*ii*)	6
(*c*)	$n_2 = 6$ to $n_1 = 2$	(*iii*)	15
(*d*)	$n_2 = 6$ to $n_1 = 1$	(*iv*)	8

(*a*) (*a*)-(*iv*), (*b*)-(*i*), (*c*)-(*iii*), (*d*)-(*ii*)
(*b*) (*a*)-(*ii*), (*b*)-(*ii*), (*c*)-(*i*), (*d*)-(*iii*)
(*c*) (*a*)-(*ii*), (*b*)-(*iii*), (*c*)-(*i*), (*d*)-(*iv*)
(*d*) (*a*)-(*iii*), (*b*)-(*ii*), (*c*)-(*iv*), (*d*)-(*i*)

61. If 13.6 eV energy is required to separate a hydrogen atom into a proton an electron. Now match the following:

Column I		Column II	
(*a*)	Total energy of electron (joules)	(*i*)	$= 4.4 \times 10^{-18}$
(*b*)	Velocity of electron (m/s)	(*ii*)	2.2×10^{6}
(*c*)	Potential energy of electron joules)	(*iii*)	5.3×10^{-11}
(*d*)	Radius of metre (orbit)	(*iv*)	-2.2×10^{-18}

(*a*) (*a*)-(*ii*), (*b*)-(*iii*), (*c*)-(*iv*), (*d*)-(*i*)
(*b*) (*a*)-(*iv*), (*b*)-(*iii*), (*c*)-(*ii*), (*d*)-(*i*)
(*c*) (*a*)-(*i*), (*b*)-(*iii*), (*c*)-(*ii*), (*d*)-(*iv*)
(*d*) (*a*)-(*iv*), (*b*)-(*ii*), (*c*)-(*i*), (*d*)-(*iii*)

62. Two electrons are in orbit if radii r_0 and $4r_0$ in an hydrogen atom. Their frequencies of revolution around the nucleus are in the ratio

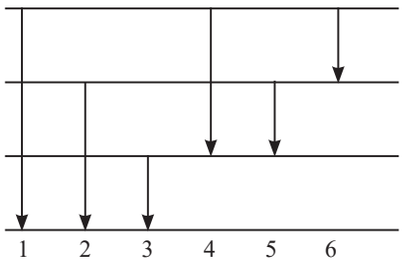
(*a*) 1 : 8 (*b*) 1 : 4 (*c*) 8 : 1 (*d*) 4 : 1

63. E_n and J_n denote the total energy and the angular momentum of an electron in the nth allowed orbit of a Bohr atom. Choose the correct relation between E_n and J_n

(*a*) $E_n \propto J_n$ (*b*) $E_n \propto 1/J_n$
(*c*) $E_n \propto J_n^2$ (*d*) $E_n \propto 1/J_n^2$

Absorption spectrum

64. The figure indicates that energy level diagram of an atom and the origin of six spectral lines in emission (*e.g.*, line number 5 arises from the transition from level B to A). Which of the following spectral lines will also occur in the absorbtion spectrum?



(*a*) 4, 5, 6 (*b*) 1, 2, 3, 4, 5, 6
(*c*) 1, 2, 3 (*d*) 1, 4, 6

Ionization energy

65. The ground state energy of H-atom is – 13.6 eV. The energy needed to ionize H-atom from its second excited state

(*a*) 1.51 eV (*b*) 13.6 eV

(*c*) 3.4 eV (*d*) None of these

66. A He^+ ion is in its first excited state. Its ionization energy is:

(*a*) 6.04 eV (*b*) 13.60 eV

(*c*) 54.40 eV (*d*) 48.36 eV

67. If the potential energy of the electron in its first allowed orbit in hydrogen atom is *E*, its

(*a*) Ionisation potential energy is – *E*/2

(*b*) Kinetic energy is – *E*/2

(*c*) Total energy *E*/2

(*d*) All of the above statements are is true

Answer Key

1. (*a*)	**2.** (*c*)	**3.** (*d*)	**4.** (*a*)	**5.** (*c*)	**6.** (*c*)	**7.** (*a*)	**8.** (*a*)	**9.** (*a*)	**10.** (*b*)
11. (*c*)	**12.** (*d*)	**13.** (*c*)	**14.** (*c*)	**15.** (*c*)	**16.** (*d*)	**17.** (*b*)	**18.** (*a*)	**19.** (*a*)	**20.** (*a*)
21. (*a*)	**22.** (*a*)	**23.** (*a*)	**24.** (*a*)	**25.** (*c*)	**26.** (2)	**27.** 4.25Å	**28.** (*a*)	**29.** (*b*)	**30.** (*a*)
31. (*d*)	**32.** (*b*)	**33.** (*b*)	**34.** (*c*)	**35.** (*d*)	**36.** (*c*)	**37.** (*d*)	**38.** (*b*)	**39.** (*b*)	**40.** (*c*)
41. (*a*)	**42.** (*a*)	**43.** (*b*)	**44.** (*d*)	**45.** (144)	**46.** (*d*)	**49.** (*d*)	**48.** (*d*)	**49.** (*a*)	**50.** (*d*)
51. (*d*)	**52.** (*c*)	**53.** (*c*)	**54.** (*b*)	**55.** (*a*)	**56.** (*d*)	**57.** (*d*)	**58.** 122	**59.** (5)	**60.** (*b*)
61. (*d*)	**62.** (*c*)	**63.** (*d*)	**64.** (*c*)	**65.** (*a*)	**66.** (*b*)	**67.** (*d*)			